Digi Pharm

R1 POC – Data Maturity

Cluster R1 End of POC Report: Data Maturity Index

Introduction

The contract management KPI framework for Cluster R1 of Saudi Arabia, a healthcare cluster comprising several hospitals and primary care facilities, currently straddles the '**managed'** to **'standardised'** level on the data maturity index. At this stage, the cluster has established standards for data management, and has ensured (broadly) consistent data quality, and has started thinking about using data for decision-making, paving the way to enhancing the quality of healthcare delivery. As we move forward into 2024-25, the aim is to evolve Cluster R1's data maturity level from 'standardised' to 'optimised.'

Immediate Plans (2023-24)

Whilst significant strides have been made in advancing the data maturity at Cluster R1, we recognise that there are still elements of the contract management KPI framework that are operating at a 'managed' level of maturity. This indicates that practices in these areas, although formal and with assigned governance, could benefit from more standardisation and efficiency. Therefore, our immediate focus will be to elevate these aspects to a 'standardised' model of delivery. This transition involves implementing uniform standards for data management, improving data quality, and integrating the use of data into our decision-making processes. By targeting these areas, we can ensure consistent performance across all aspects of contract management, further enhancing the overall data maturity and setting the stage for future growth and optimisation.

2024-25

This transition towards an optimised level focuses on enhancing efficiency, with the goal of leveraging data for timely insights that can be used to make proactive, informed decisions. To achieve this, we plan on implementing a highly standardised and automated reporting model.

The reporting model will streamline the data collection process, ensuring data from various sources within the healthcare cluster is consolidated quickly and accurately. It

will reduce manual intervention, minimising errors, and freeing up valuable resources. The automated reporting system will enhance real-time visibility into healthcare operations, providing decision-makers with the necessary data at their fingertips.

Once this efficient reporting model is established, our attention will turn to predictive analytics. The ability to forecast future trends is a significant aspect of the 'optimised' level on the data maturity index. By using statistical methods, such as linear regression, we will analyse current and historical facts to make predictions about future or otherwise unknown events.

In terms of tools to conduct this analysis, we will use R, Python and other statistical tools.

In practice, this could mean predicting patient influx during specific seasons, foreseeing potential health crises, or anticipating resource needs. These predictive insights will allow Cluster R1 to be proactive rather than reactive, enhancing resource planning and improving patient care.

2025 and Beyond

Once these steps are taken and Cluster R1 has comfortably settled into the 'optimised' level of data maturity, the next aim is to venture into the 'innovating' stage. Here, we envision leveraging advanced analytics, machine learning, and artificial intelligence to uncover new insights, predict patient outcomes, personalise treatment plans, and truly drive innovation in healthcare delivery.

Thus, the journey for Cluster R1 from 'standardised' to 'optimised' and then 'innovating' involves enhancing efficiency, embracing predictive analytics, and eventually integrating advanced data techniques, thereby transforming healthcare delivery within this cluster.